



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to SANTHOSH AGENCIES

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

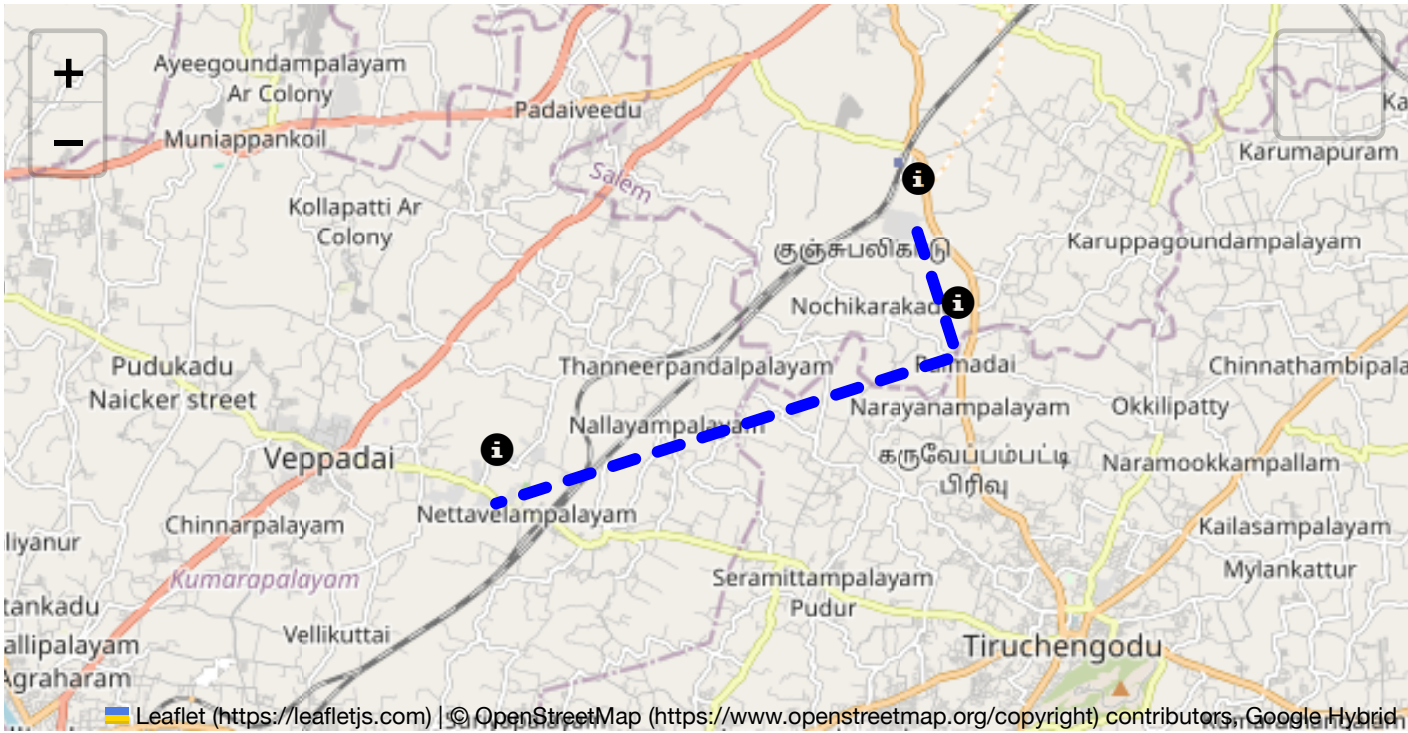
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

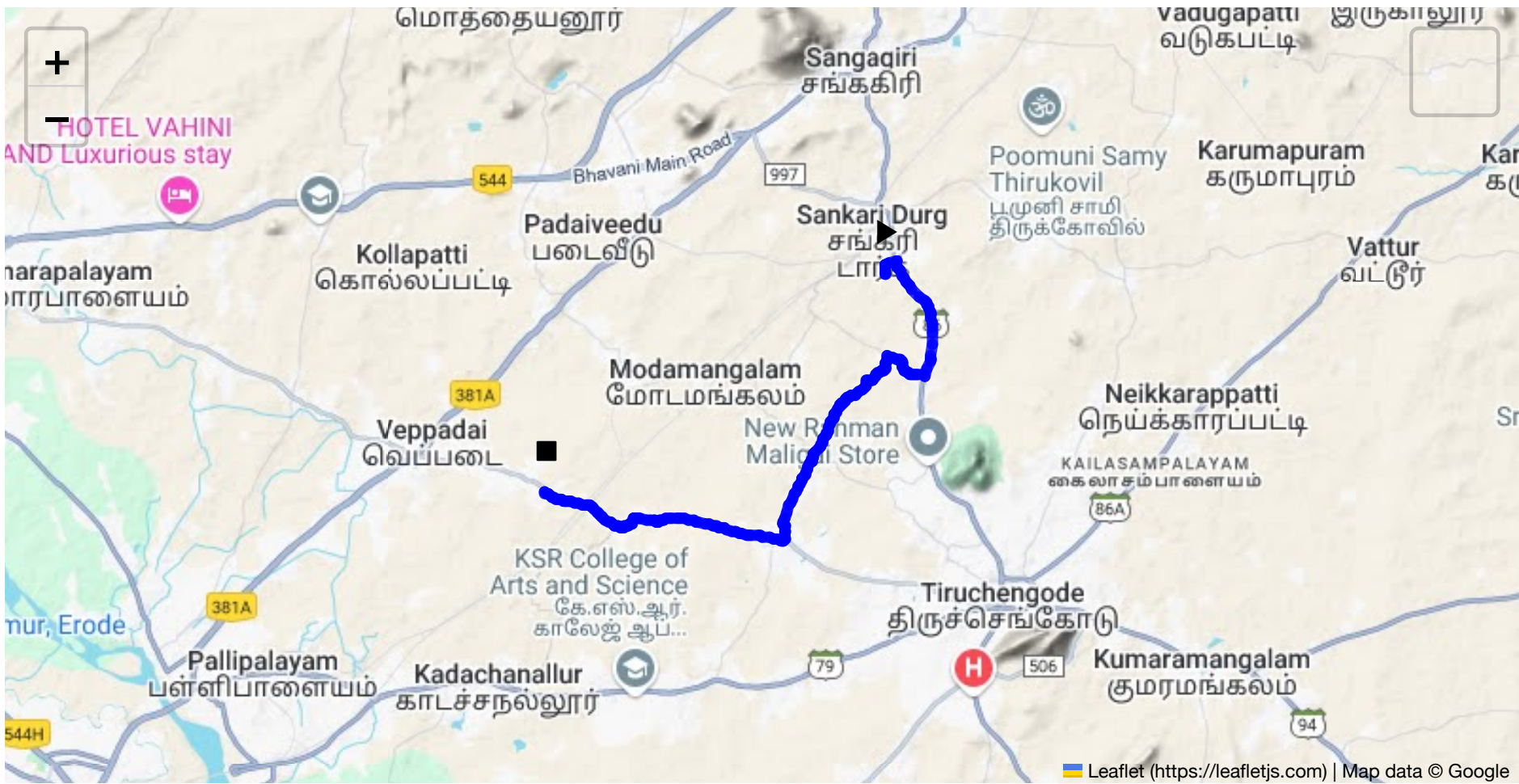
Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 13.49 km
Estimated Duration: 0.4 hours
Adjusted Duration (Heavy Vehicle): 0.5 hours

Start: (11.4381, 77.8734)
End: (11.399071, 77.811562)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map:
 - Starting from Sangagiri (CVQF+23W), the route proceeds southeast towards Palmadai (CVCH+7V8) and then continues to Nettavelampalayam (9RX6+JJ). The total distance is approximately 13.49 kilometers.
 - The route primarily utilizes regional roads, passing through rural and semi-urban areas.
2. Typical Weather Conditions and Potential Weather-Related Hazards:
 - Weather:** Generally hot and humid, with monsoon seasons bringing heavy rainfall.
 - Hazards:** Monsoon rains can lead to waterlogging and slippery roads. Occasional fog during winter mornings can reduce visibility.
3. Analysis of Traffic Patterns:
 - Peak Hours:** Morning (7:30 - 9:30 AM) and evening (5:00 - 7:00 PM).
 - Congestion-Prone Areas:** The intersection points near towns and market areas can get congested, especially during peak hours.
4. Assessment of Road Quality and Infrastructure:
 - Road Quality:** Varies from good in some stretches to average; some sections may have potholes or uneven surfaces, especially after rains.
 - Infrastructure:** Basic infrastructure with occasional street lighting; signage might be sparse but adequate for navigation.
5. Suggestions for Alternative Routes for Emergencies:
 - If the main route is blocked, consider taking local roads that bypass congested areas. Consult local maps for alternative rural pathways, which may not be well-paved but can provide access.
6. Summary of Local Regulations Affecting Hazardous Material Transport:

- Compliance with national hazardous material regulations is mandatory, including adhering to weight limits and obtaining necessary permits.
- Certain materials might have restricted access during peak traffic hours to avoid congestion.

7. **Overview of Historical Incidents Involving Heavy Vehicles or Hazardous Materials:**

- There have been rare incidents of vehicle overturns due to sharp turns or road quality issues.
- Ensure proper load securing and adherence to speed limits to mitigate such risks.

8. **Environmental Considerations and Sensitive Areas:**

- The route passes through agricultural lands; contamination from spills would affect local ecology.
- Efforts should be made to minimize noise and emissions, especially near populated areas.

9. **Analysis of Communication Coverage:**

- Generally good mobile coverage throughout most of the route, but occasional dead zones may exist in rural stretches, particularly in areas surrounded by hills.

10. **Estimated Emergency Response Times for Different Route Segments:**

- Urban Areas: Approximately 15-20 minutes due to closer proximity to emergency services.
- Rural Segments: Can range from 30 to 45 minutes depending on accessibility.

11. **Overall Summary of Risk Assessment:**

- **Risks:** The primary risks involve road quality, traffic congestion, and weather conditions, especially during the monsoon season.
- **Safety Recommendations:** Conduct regular vehicle checks, maintain communication with control hubs, and ensure all drivers are trained in emergency protocols.
- **Mitigation Measures:** Utilize safety equipment, follow speed regulations, and stay updated on weather forecasts to adjust travel plans accordingly.

Overall, while the route has some challenges, adherence to safety protocols and appropriate preparedness can mitigate most risks associated with transport of hazardous materials.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	High	11.43811, 77.87348	15 KM/Hr
1	Turn	Medium	11.43956, 77.87340	30 KM/Hr
2	Turn	Medium	11.43971, 77.87348	30 KM/Hr
3	Blind Spot	Blind Spot	11.44029, 77.87544	10 KM/Hr
4	Blind Spot	Blind Spot	11.41981, 77.88078	10 KM/Hr
5	Turn	Medium	11.42039, 77.87758	30 KM/Hr
6	Turn	Medium	11.42102, 77.87680	30 KM/Hr
7	Turn	Medium	11.42231, 77.87666	30 KM/Hr
8	Blind Spot	Blind Spot	11.42333, 77.87394	10 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
9	Turn	Medium	11.42140, 77.87364	30 KM/Hr
10	Turn	High	11.41917, 77.87022	15 KM/Hr
11	Turn	High	11.40478, 77.86020	15 KM/Hr
12	Turn	High	11.40471, 77.86004	15 KM/Hr
13	Turn	Medium	11.39361, 77.85502	30 KM/Hr
14	Turn	Medium	11.39323, 77.85539	30 KM/Hr
15	Blind Spot	Blind Spot	11.39047, 77.85509	10 KM/Hr
16	Turn	Medium	11.39431, 77.83265	30 KM/Hr
17	Turn	Medium	11.39383, 77.83222	30 KM/Hr
18	Turn	Medium	11.39381, 77.83208	30 KM/Hr
19	Turn	Medium	11.39468, 77.82800	30 KM/Hr
20	Turn	Medium	11.39692, 77.81974	30 KM/Hr
21	Turn	High	11.39915, 77.81187	15 KM/Hr

Route Photos of Risky Spots



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.44029, 77.87544



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.41981, 77.88078



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.42333, 77.87394



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.41917, 77.87022



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.40478, 77.86020



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.40471, 77.86004



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.39361, 77.85502



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.39323, 77.85539



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.39047, 77.85509



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.39431, 77.83265



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.39383, 77.83222



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.39381, 77.83208



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.39468, 77.82800



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.39692, 77.81974



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.39915, 77.81187

